SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – MCQ Test

Course Code:	CSA0305	**Course Title:**	Data Structures
CO Assessed:	CO3 – Precision (BL3, BL5)	**Assessment:**	Assessment Tool 3 – MCQ Test
Weightage:	30% of CIA	**Total Marks:**	100
CO3 Statement:	*Solve problems for optimized data storage and retrieval using Binary Search Trees, AVL Trees, B-Trees, Red-Black Trees, and Splay Trees.*		
Student Name:	K Suthesika	**Reg. No.:**	192511070
Section / Batch:	BE CSE	**Date:**	12/08/2026

Code of Conduct

*I ,*K Suthesika *(*192511070 *) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: K Suthesika

Instructions

- This test contains 25 MCQ Test questions (4 marks each), Total: 100 Marks.
- When a student answer using MCQ, in evaluation the answer should be with following requirements: Identify the most appropriate answer, conceptual understanding, problem-solving ability, application of knowledge.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Preliminaries – Tree Terminology	Root, height, level, siblings and sub tree	1–2	8
Binary Tree, Binary Search Tree	Properties, binary tree and binary search tree, insertion and deletion	3–9	28
Tree Traversals	Traversal types, In order, Post order and Pre order	10–14	20
AVL Tree	Balancing factor, Rotation, Types of rotation, examples	15–16	8
Applications of Search Trees	Usage of binary tree	17	4
B Tree - 2-3 tree, 2-3-4 tree	Properties, Example and Operations	18–19	8
TRIE	Properties, structures and Operations	20	4
Red Black Tree	Properties, Balancing and Operations	21–22	8
Splay Tree	Splaying, Types and examples for insertion and deletion	23–25	12
		GRAND TOTAL:	**100 Marks**

MCQ

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Conceptual Understanding	20	Demonstrates thorough understanding of all core Data Structures concepts; answers accurately reflect deep knowledge	Shows good understanding with minor conceptual gaps	Basic understanding; several misconceptions present	Limited understanding; major concepts misunderstood
Application Skills	20	Correctly applies concepts to solve complex and scenario-based MCQs	Applies concepts correctly in most moderate-level questions	Struggles with application in complex scenarios	Unable to apply concepts effectively
Analytical Thinking	30	Consistently selects correct answers for high-order questions	Performs well in moderate analytical questions	Limited success in analytical questions	Unable to interpret analytical/problem-based questions
Time Management	20	Completes test efficiently with time for review	Completes test within time	Struggles to complete test on time	Unable to complete test
Accuracy of Responses	10	90–100% correct answers	75–89% correct answers	50–74% correct answers	Below 50% correct answers

Faculty In-charge	Course Coordinator	Program Director
Signature: _____________	Signature: _____________	Signature: _____________
Date: ________________	Date: ________________	Date: ________________

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Tree Traversal
4 marks

Q2
Binary Search Tree
4 marks

Q3
Red-Black Tree
4 marks

Q4
Red-Black Tree
4 marks

Q5
B-Tree
4 marks

Q6
Tree Terminology
4 marks

Q1 **Tree Traversal** 4 marks

You analyze the post-order traversal of a tree representing project tasks: [D, E, B, F, G, C, A]. You want to know

A. 3

B. 4

C. 5

D. 2

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Q1
Tree Traversal
4 marks

Q2
Binary Search Tree
4 marks

Q3
Red-Black Tree
4 marks

Q4
Red-Black Tree
4 marks

Q5
B-Tree
4 marks

Q6
Tree Terminology
4 marks

Q2 **Binary Search Tree** 4 marks

In a Binary Search Tree, the left child contains:

A. Greater values

B. Smaller values

C. Equal values

D. Random values

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- Q2 Binary Search Tree 4 marks
- Q3 Red-Black Tree 4 marks
- Q4 Red-Black Tree 4 marks
- Q5 B-Tree 4 marks
- Q6 Tree Terminology 4 marks

Q3 **Red-Black Tree** 4 marks

During insertion, when a red node has a red parent and the uncle is also red, what action is taken:

A. Single rotation

B. Double rotation

C. Recolor parent and uncle black, grandparent red

D. No action needed

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- Q4 Red-Black Tree 4 marks
- Q5 B-Tree 4 marks
- Q6 Tree Terminology 4 marks

Q4 **Red-Black Tree** 4 marks

Which property ensures that no two red nodes are adjacent in a Red-Black Tree?

A. Root is black

B. Red nodes have black children

C. Every path from root to leaf has the same number of black nodes

D. Leaves are red

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- Q4 Red-Black Tree 4 marks
- Q5 B-Tree 4 marks
- Q6 Tree Terminology 4 marks

Q5 **B-Tree** 4 marks

A 2-3-4 tree contains a 4-node with keys [20, 40, 60]. If a new key 50 is to be inserted, what is the result of the split?

A. New root becomes [40]; children: [20], [50], [60]

B. Root stays [20, 40, 50, 60]

C. Keys rearranged into a single 4-node

D. Insertion is not possible

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- Q4 Red-Black Tree 4 marks
- Q5 B-Tree 4 marks
- Q6 Tree Terminology 4 marks

Q6 **Tree Terminology** 4 marks

A subtree of a tree is:

A. A separate graph

B. A part of the tree that is itself a tree

C. Only leaf nodes

D. Only internal nodes

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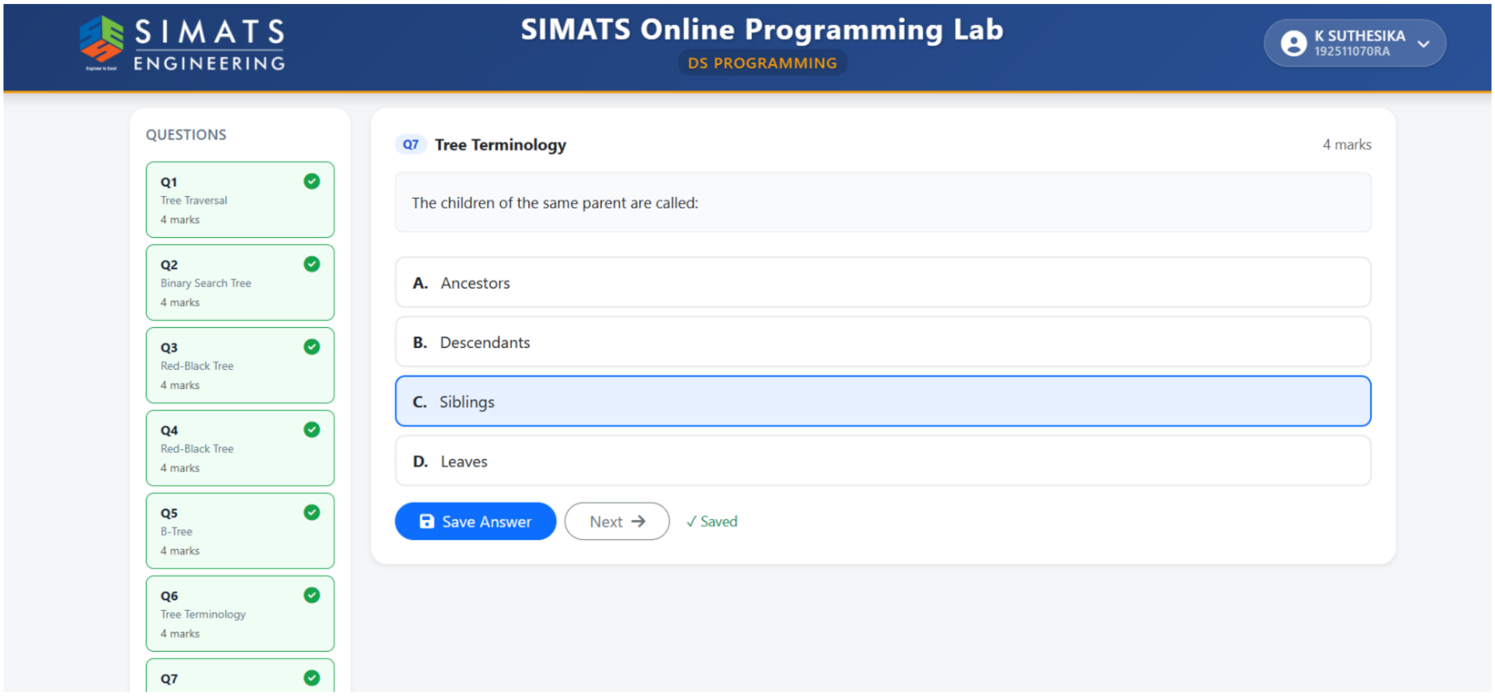
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4 marks
Q5
B-Tree
4 marks
Q6
Tree Terminology
4 marks
Q7
Q7 Tree Terminology
4 marks
The children of the same parent are called:
A. Ancestors
B. Descendants
C. Siblings
D. Leaves
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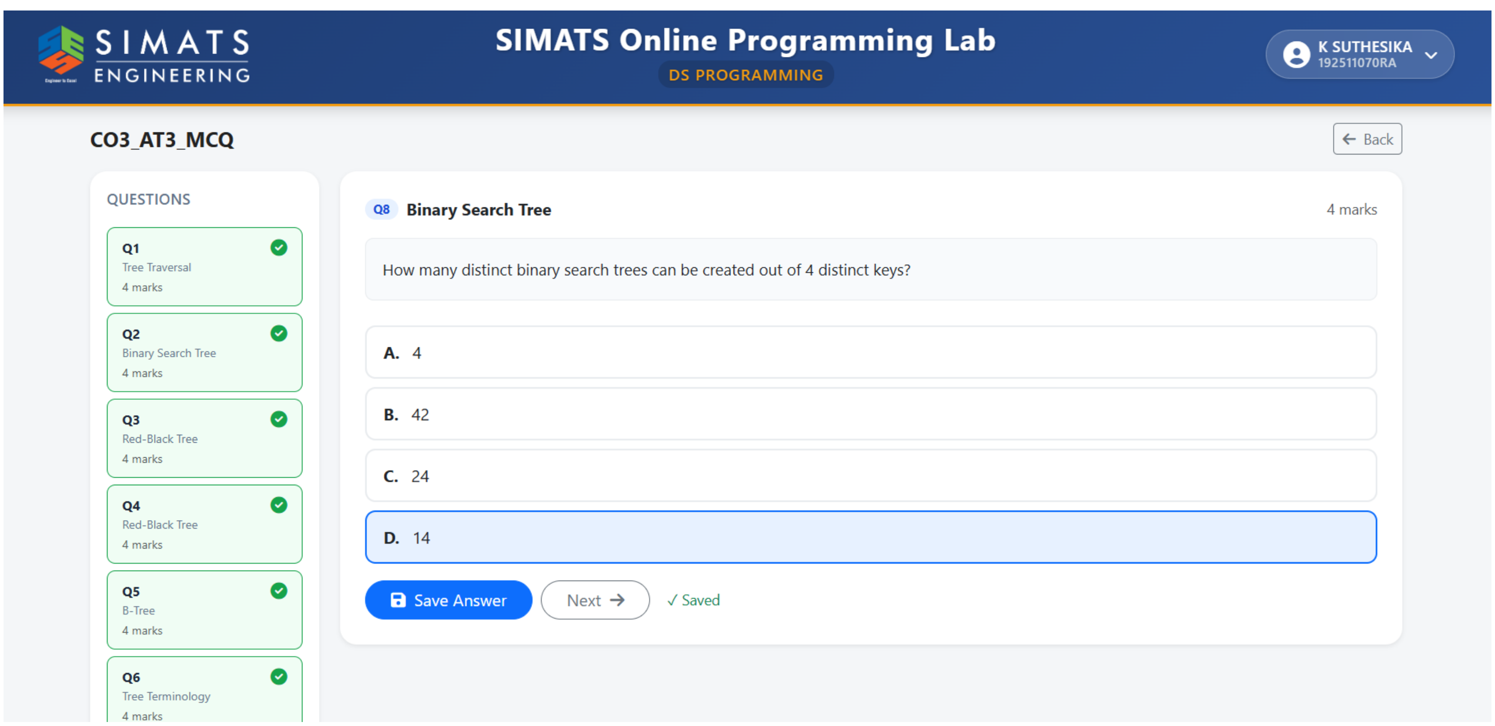
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4 marks
Q4
Red-Black Tree
4 marks
Q5
B-Tree
4 marks
Q6
Tree Terminology
4 marks
Q8 Binary Search Tree
4 marks
How many distinct binary search trees can be created out of 4 distinct keys?
A. 4
B. 42
C. 24
D. 14
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- Q3 Red-Black Tree 4 marks
- Q4 Red-Black Tree 4 marks
- Q5 B-Tree 4 marks
- Q6 Tree Terminology

Q9 **Binary Tree** 4 marks

The left and right subtrees of a binary tree are themselves:

A. Trees

B. Graphs

C. Arrays

D. Linked Lists

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- Q4 Red-Black Tree 4 marks
- Q5 B-Tree 4 marks
- Q6 Tree Terminology 4 marks

Q10 **Binary Tree** 4 marks

A complete binary tree is one in which:

A. All levels are fully filled

B. Every node has two children

C. All levels except possibly the last are completely filled

D. Only root is filled

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- Q4 Red-Black Tree 4 marks
- Q5 B-Tree 4 marks
- Q6 Tree Terminology 4 marks

Q11 **Binary Tree** 4 marks

The maximum number of nodes at level 'i' in a binary tree is:

A. 2^i

B. i

C. i^2

D. 2i

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- Q4 Red-Black Tree 4 marks
- Q5 B-Tree 4 marks
- Q6 Tree Terminology 4 marks

Q12 **Binary Search Tree** 4 marks

In a BST managing product SKUs, values 20, 15, 25, 10, 18, 30 are inserted. To optimize retrieval, you check the number of edges between node 10 and the root. How many edges are in the path from node 10 to the root?

A. 1

B. 3

C. 2

D. 5

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- Q4 Red-Black Tree 4 marks
- Q5 B-Tree 4 marks
- Q6 Tree Terminology 4 marks

Q13 **Tree Traversals** 4 marks

You are debugging a program that reconstructs the original root of a binary decision tree using traversal logs: In-order and post-order. Given: In-order: D B E A F C Post-order: D E B F C A What is the root

A. A

B. D

C. B

D. C

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- Q4 Red-Black Tree 4 marks
- Q5 B-Tree 4 marks
- Q6 Tree Terminology 4 marks

Q14 **Tree Traversals** 4 marks

Which traversal follows Left → Root → Right?
a)
b)
c)
d)

A. Level order

B. Preorder

C. Postorder

D. Inorder

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Red-Black Tree
4 marks

Q5
B-Tree
4 marks

Q6
Tree Terminology
4 marks

Q15 **Tree Traversals** 4 marks

Preorder traversal of an expression tree gives

A. Prefix expression

B. Infix expression

C. Postfix expression

D. Sorted expression

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4 marks

Q4
Red-Black Tree
4 marks

Q5
B-Tree
4 marks

Q6
Tree Terminology
4 marks

Q7
Tree Terminology

Q16 **Tree Traversals** 4 marks

Consider the binary tree:
10
/ \
20 30
/ \ / \
40 50 60 70
What is the Preorder traversal?

A. 40 20 50 10 60 30 70

B. 40 50 20 60 70 30 10

C. 10 20 40 50 30 60 70

D. 10 30 60 70 20 40 50

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Red-Black Tree
4 marks

Q5
B-Tree
4 marks

Q6
Tree Terminology
4 marks

Q7
Tree Terminology
4 marks

Q17 **AVL Tree** 4 marks

The balance factor of an AVL tree node can be:

A. -1, 0, +1

B. -2, -1, 0

C. 0, 1, 2

D. Any value

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Q4
Red-Black Tree
4 marks

Q5
B-Tree
4 marks

Q6
Tree Terminology
4 marks

Q7
Tree Terminology
4 marks

Q18 **AVL Tree** 4 marks

Consider the AVL tree: What is the balance factor of node 30?

30

/ \

20 40

/ \

10 25

B. +1

C. -1

D. +2

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- Q4 Red-Black Tree 4 marks
- Q5 B-Tree 4 marks
- Q6 Tree Terminology 4 marks

Q19 **Search Tree** 4 marks

Which of the following is a common application of Binary Search Trees (BST)?

A. Database indexing

B. Video editing

C. Audio compression

D. Network wiring

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- Q4 Red-Black Tree 4 marks
- Q5 B-Tree 4 marks
- Q6 Tree Terminology 4 marks

Q20 **B-Tree** 4 marks

In a B-Tree of order 4, what is the maximum number of children a node can have?

A. 2

B. 3

C. 4

D. 5

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Q2
Binary Search Tree
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Q3
Red-Black Tree
4 marks

Q4
Red-Black Tree
4 marks

Q5
B-Tree
4 marks

Q6
Tree Terminology
4 marks

Q21 **Trie**

4 marks

In Trie, each node represents:

A. Prefix matching

B. Arithmetic operations

C. Graph traversal

D. Sorting arrays

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Red-Black Tree
4 marks

Q4
Red-Black Tree
4 marks

Q5
B-Tree
4 marks

Q6
Tree Terminology
4 marks

Q7
Tree Terminology
4 marks

Q22 **Splay Tree**

4 marks

Splay Tree is a type of:

A. Complete Binary Tree

B. Self-adjusting Binary Search Tree

C. Heap Tree

D. Graph

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4 marks

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Red-Black Tree
4 marks

Q4
Red-Black Tree
4 marks

Q5
B-Tree
4 marks

Q6
Tree Terminology
4 marks

Q23 **Tree** 4 marks

The height of a single-node tree is:

B. 1

C. -1

D. 2

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Red-Black Tree
4 marks

Q4
Red-Black Tree
4 marks

Q5
B-Tree
4 marks

Q6
Tree Terminology
4 marks

Q7

Q24 **Splay Tree** 4 marks

Which rotation is used when the node is the direct child of the root?
20
/
10

A. Zig

B. Zig-Zig

C. Zag-Zig

D. Split

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Red-Black Tree
4 marks

Q4
Red-Black Tree
4 marks

Q5
B-Tree
4 marks

Q6
Tree Terminology
4 marks

Q7

Q25 **Splay Tree** 4 marks

After accessing a node in a Splay Tree, that node is moved to:

A. Leaf

B. Middle

C. Root

D. NULL

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